Fields and Waves I

Lecture 8
Fields & Math I

Alex Patterson
Electrical, Computer, and Systems Engineering Department
Rensselaer Polytechnic Institute, Troy, NY

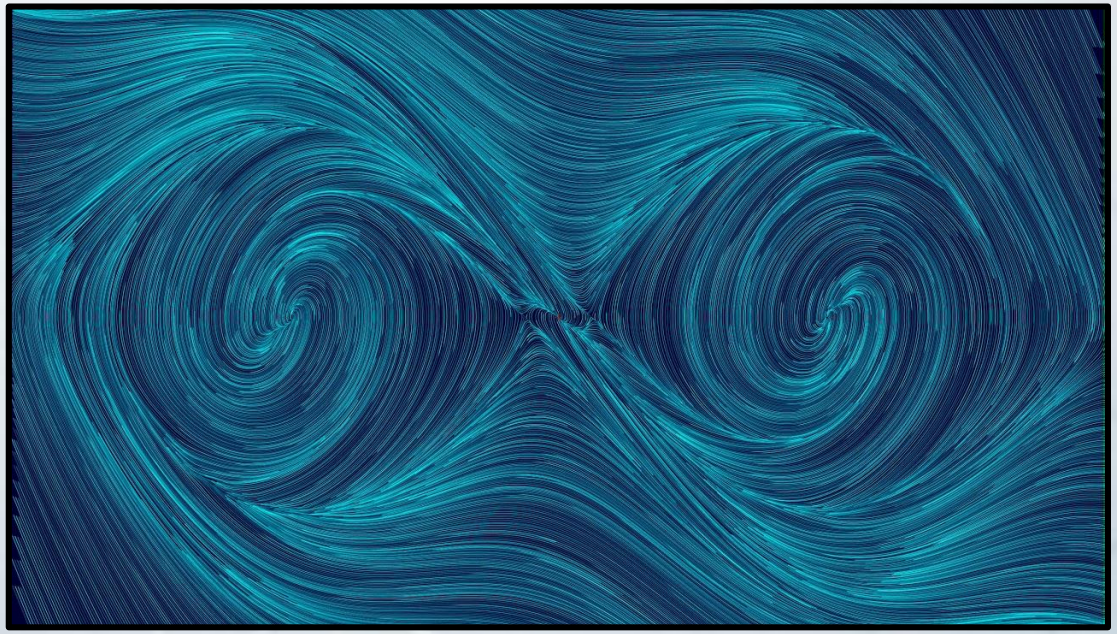
These slides were prepared through the work of the following people:

- Kenneth A. Connor – ECSE Department, Rensselaer Polytechnic Institute, Troy, NY (Emeritus)
- J. Darryl Michael – GE Global Research Center, Niskayuna, NY
- Thomas P. Crowley – National Institute of Standards and Technology, Boulder, CO
- Sheppard J. Salon – ECSE Department, Rensselaer Polytechnic Institute, Troy, NY (Emeritus)
- Lale Ergene – ITU Informatics Institute, Istanbul, Turkey
- Jeffrey Braunstein – ECE Department, University at Albany
- James Lu - ECSE Department, Rensselaer Polytechnic Institute, Troy, NY
- James Dylan Rees - ECSE Department, Rensselaer Polytechnic Institute, Troy, NY
- Alex Patterson - ECSE Department, Rensselaer Polytechnic Institute, Troy, NY

Materials from other sources are referenced where they are used.
Those listed as Ulaby are figures from Ulaby's textbook.

Overview

- Orthogonal Coordinate Systems
- Multidimensional Integrals



Field Math

- In our treatment of transmission lines so far, we have been able to derive voltage and current everywhere on the line using an extension of circuit theory.

- But to understand the more general picture, we need Maxwell's Equations. **Now we enter the "fields" portion of the class.**

$\nabla . E = \frac{\rho}{\epsilon_0}$	(1)	Gauss' law
$\nabla . B = 0$	(2)	Magnetic monopoles
$\nabla \times E = -\frac{\partial B}{\partial t}$	(3)	Faraday's law
$\nabla \times H = J + \frac{\partial D}{\partial t}$	(4)	Ampere-Maxwell law

Field Math

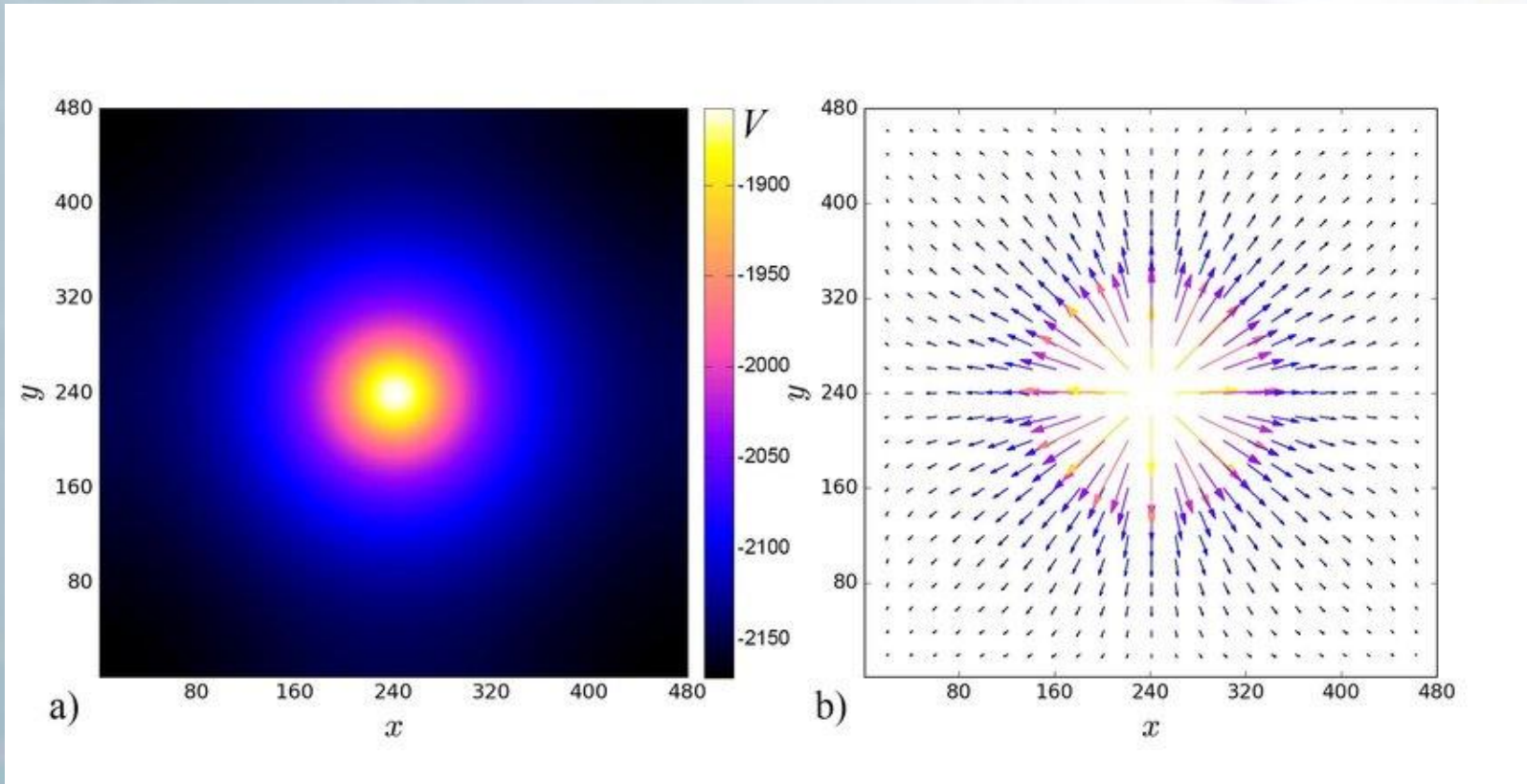
...but what is a field, anyway?

Field Math

...but what is a field, anyway? A field:

- Has a value at every point in space
- Could be a scalar field, where every point has scalar quantity (i.e. electric potential)
- Could be a vector field, where every point has a vector (i.e. electric field, magnetic field)
- Scalar and vector fields are often related through gradients

Field Math



Electric potential field (left) and corresponding electric field (vector field, right) for a charged droplet (Lauricella)

Field Math – "Field or no Field?"

Which of these are represented mathematically by fields? Why or why not?

- Wind velocity — field – magnitude + direction – vector field
- Energy density — field – magnitude – scalar field
- Temperature — field – magnitude – scalar field
- Friction force — not a field – not defined at all points (only on contact surfaces)

Orthogonal Coordinate Systems

Field Math

- 3 Types of coordinate systems we will use for fields:
  - Rectangular
  - Cylindrical
  - Spherical
- Examples of when to use them:
  - Conductive sheets (rectangular)
  - Wires and cables (cylindrical)
  - Spheres (spherical)
- By choosing our coordinate systems for symmetry reasons, we can make the math easier. For instance, both cables and the cylindrical coordinate system have radial symmetry.

Field Math

Rectangular / Cartesian Coordinates Definition

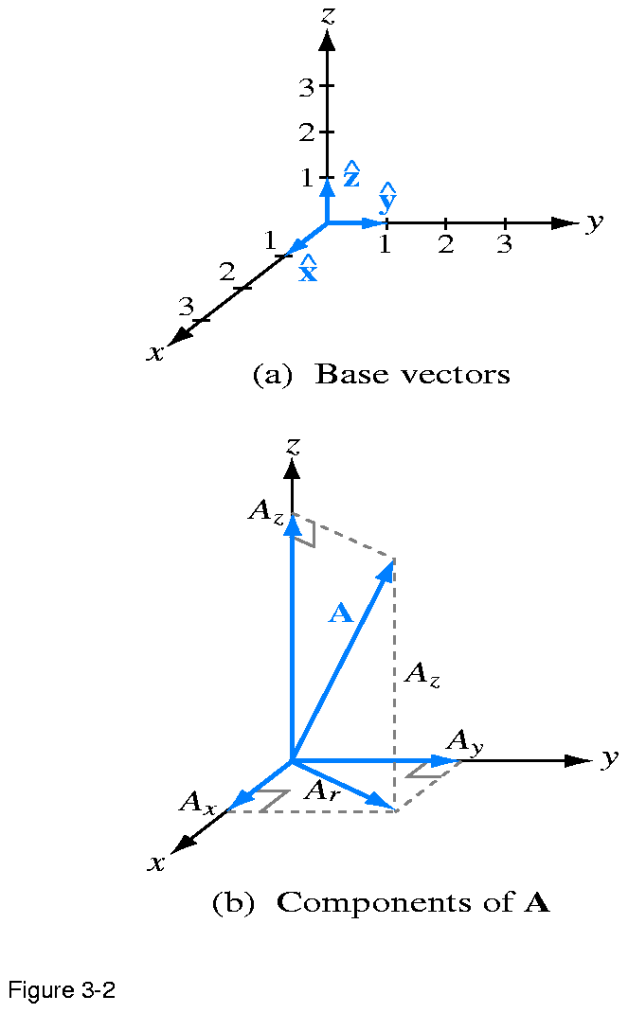
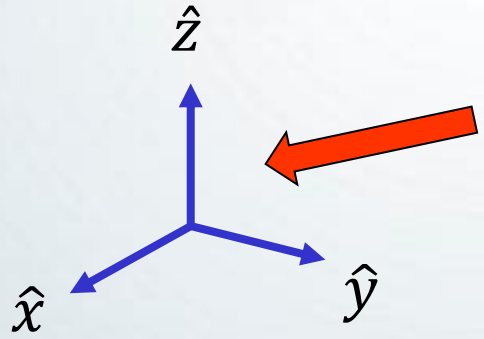


Figure 3-2

General Vector A in Rectangular Coordinates:

$$\vec{A} = A_x\hat{x} + A_y\hat{y} + A_z\hat{z}$$



Unit Vector Representation for Rectangular Coordinate System

The Unit Vectors imply:

- $\hat{x}$ ➡ Points in the direction of increasing **x**
- $\hat{y}$ ➡ Points in the direction of increasing **y**
- $\hat{z}$ ➡ Points in the direction of increasing **z**

Field Math

Rectangular / Cartesian Coordinates Dot Product

- Definition of Dot Product (Scalar Product)

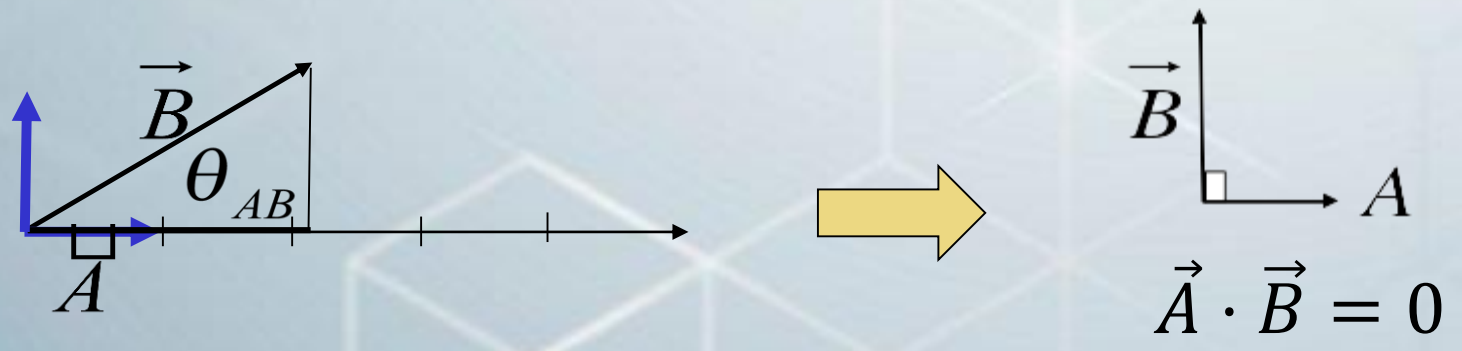
$$\vec{A} \cdot \vec{B} = \left|\vec{A}\right|\left|\vec{B}\right| \cos(\theta_{AB})$$

$$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$$

$$\left|\vec{A}\right| = \sqrt{\vec{A} \cdot \vec{A}} = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

Magnitude of vector

- Meaning of dot product

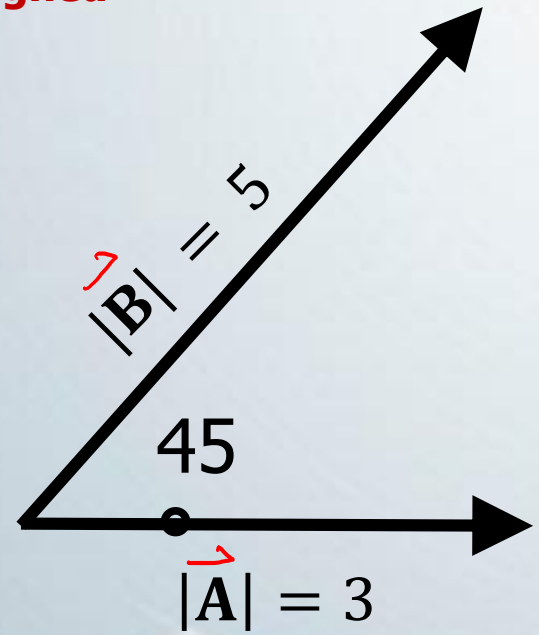


Field Math

Rectangular / Cartesian Coordinates Dot Product

- What's the dot product of **A** and **B**?

A measure of how aligned two vectors are



Note: we can denote vectors either by putting an arrow over their letters or by writing their letters in bold.

$$\vec{A} \cdot \vec{B} = \left|\vec{A}\right|\left|\vec{B}\right| \cos\theta = 3 \cdot 5 \cdot \cos 45° = 10.6$$

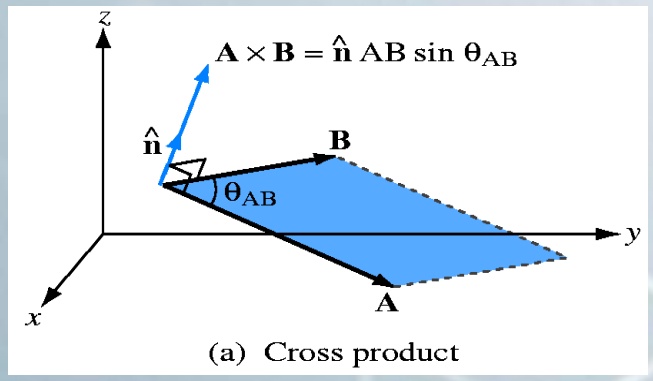
Field Math

Rectangular / Cartesian Coordinates Cross Product

- Definition of Cross Product (Vector Product)

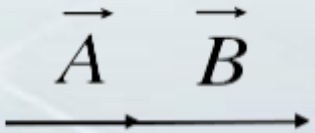
$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

- Meaning of the cross product



(a) Cross product

Gives the area of a parallelogram with sides defined by A and B

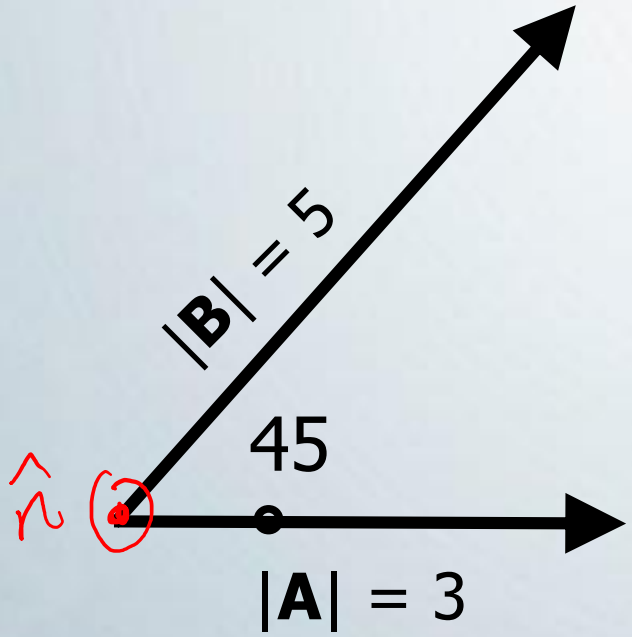


$$\vec{A} \times \vec{B} = 0$$

Field Math

Rectangular / Cartesian Coordinates Cross Product

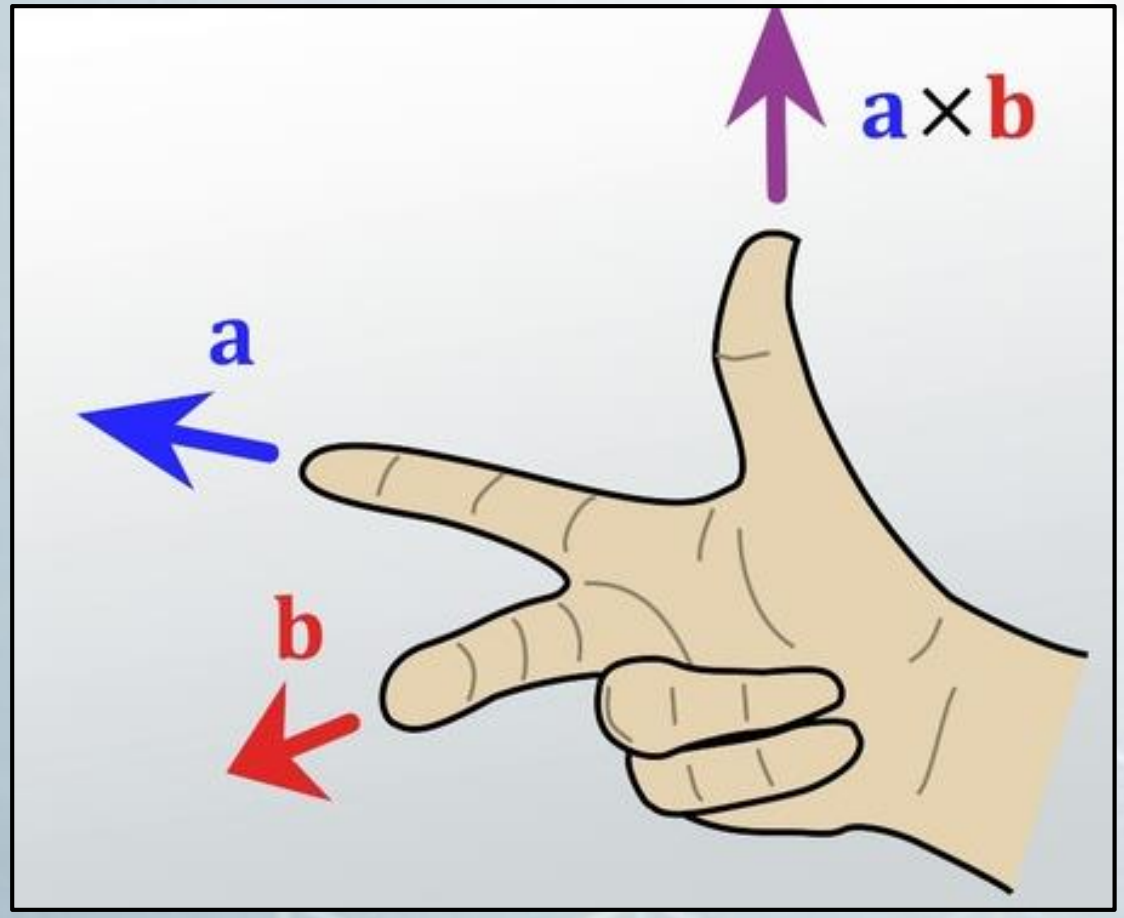
- What's the **cross product** of **A** and **B**?



$$\vec{A} \times \vec{B} = \left|\vec{A}\right|\left|\vec{B}\right| \sin\theta\, \hat{n} = 3 \cdot 5 \cdot \sin 45° \hat{n} = 10.6\hat{n}$$

Field Math

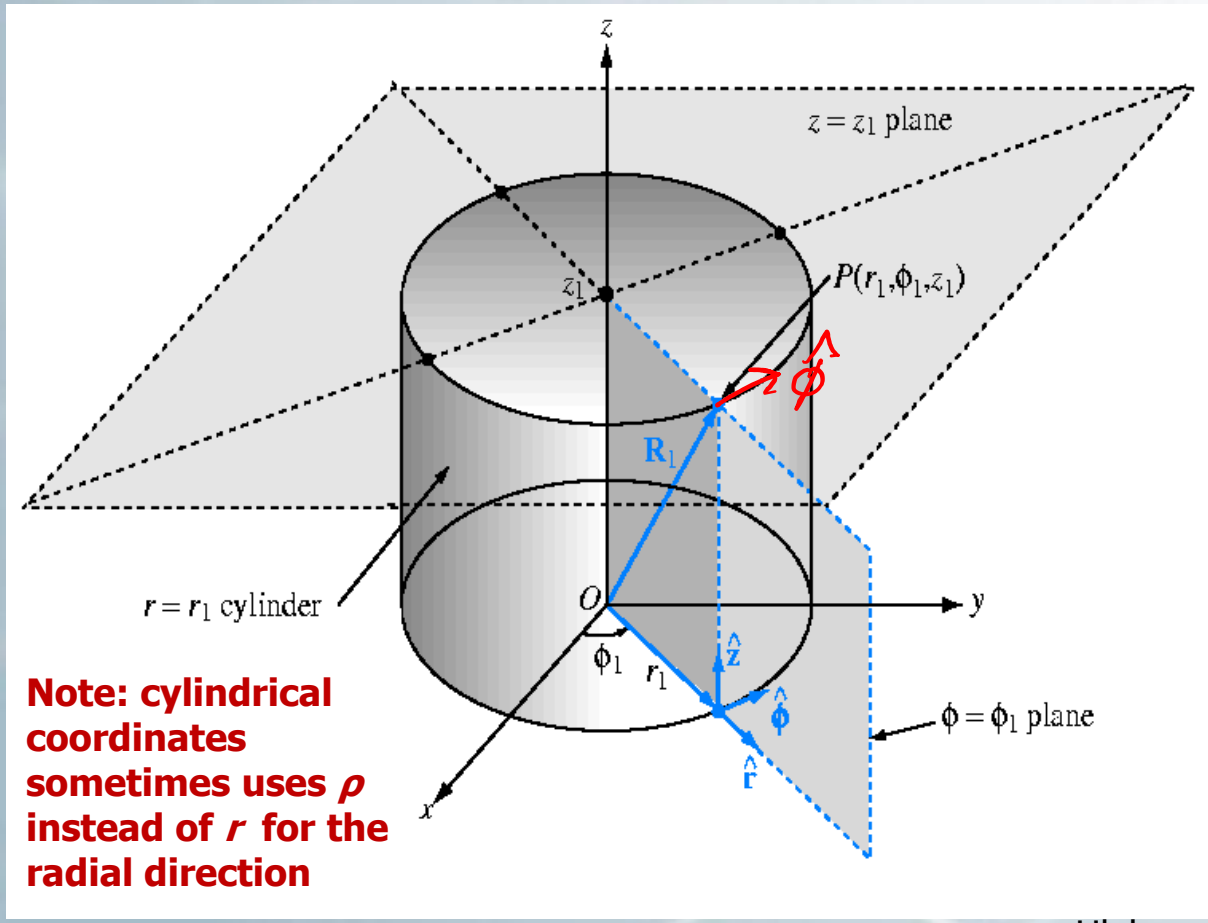
The Right Hand Rule



Study.com

Field Math

Cylindrical Coordinates Definition



Note: cylindrical coordinates sometimes uses *ρ* instead of *r* for the radial direction

Ulaby

Cylindrical Coordinates

- r: radial coordinate (length of vector in x-y plane)
- ϕ: angle between vector and x-axis in x-y plane
- z: z in Cartesian coordinates

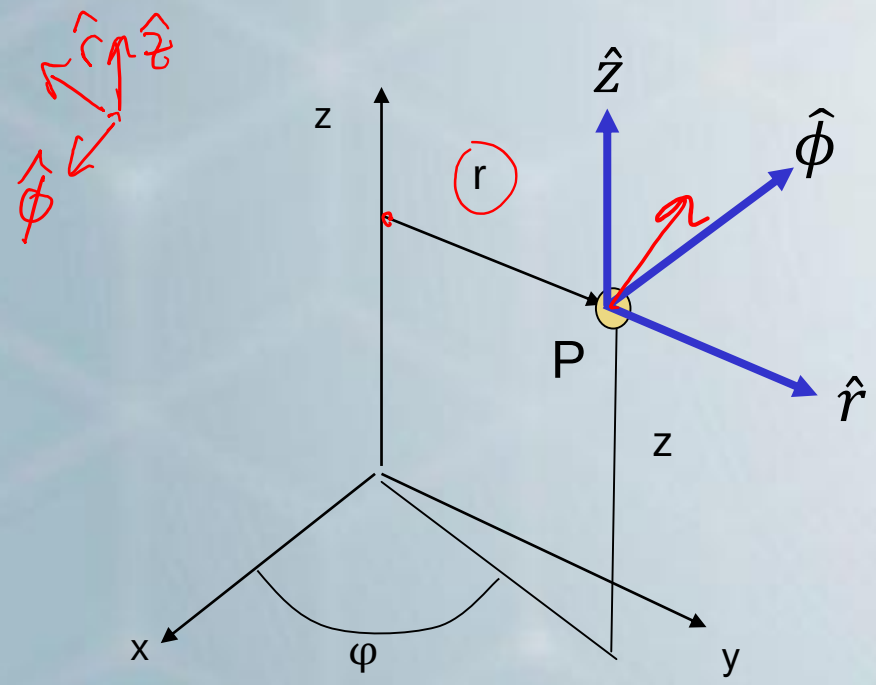
Note: there is no vector length component for φ: *r* is the length of the vector in the x-y plane and the direction of *r* is dependent on *φ*

Position Vector R in Cylindrical Coordinates:

$$\overrightarrow{OP} = \mathbf{r} = r_1\hat{\boldsymbol{r}} + z_1\hat{\boldsymbol{z}} \text{ for } P(r_1,\phi_1,z_1)$$

Field Math

Cylindrical Coordinates Unit Vector



General Vector A in Cylindrical Coordinates:

$$\vec{A} = A_r\hat{\boldsymbol{r}} + A_\phi\hat{\boldsymbol{\phi}} + A_z\hat{\boldsymbol{z}}$$

The Unit Vectors imply :

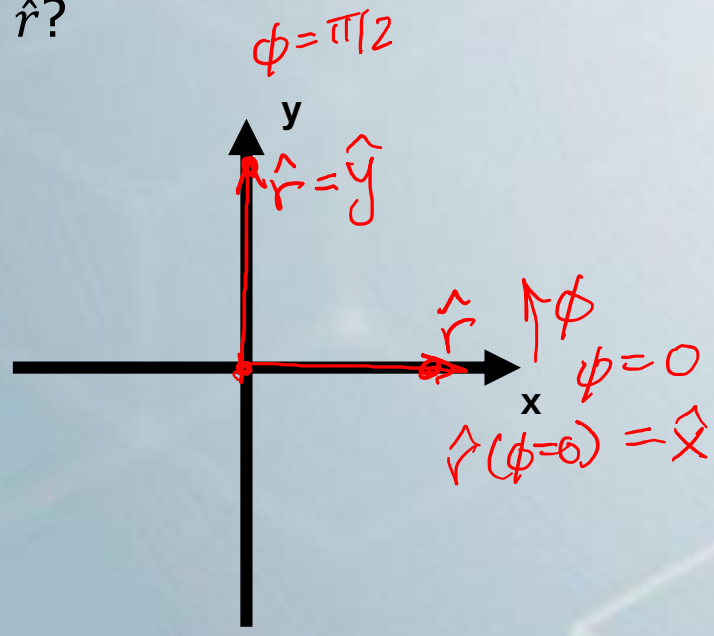
- $\hat{r}$ ➡ Points in the direction of increasing **r** (outward from 0)
- $\hat{\phi}$ ➡ Points in the direction of increasing **ϕ** (moving counter-clockwise from the +x-axis)
- $\hat{z}$ ➡ Points in the direction of increasing **z**

In cylindrical coordinates, both $\hat{r}$ and $\hat{\phi}$ are functions of ϕ

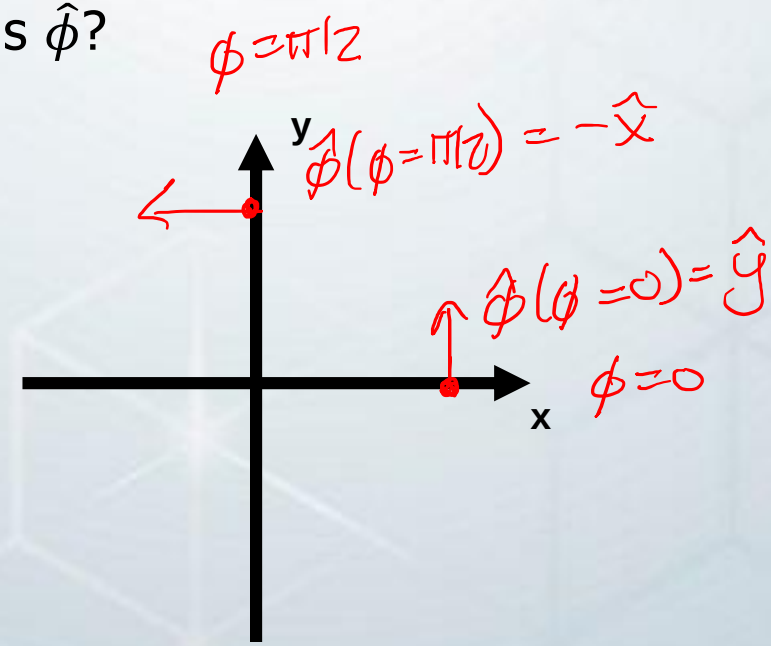
Field Math

Cylindrical Coordinates Unit Vector

- What are the unit vectors $\hat{r}$, $\hat{\phi}$, and $\hat{z}$ of cylindrical coordinates in terms of Cartesian Coordinates?
  - What is $\hat{z}$? $\hat{z}$
  - What is $\hat{r}$?



What is $\hat{\phi}$?



$$\hat{r} = (\cos\phi)\hat{x} + (\sin\phi)\hat{y}$$

$$\hat{\phi} = -(\sin\phi)\hat{x} + (\cos\phi)\hat{y}$$

Field Math

Spherical Coordinates Definition

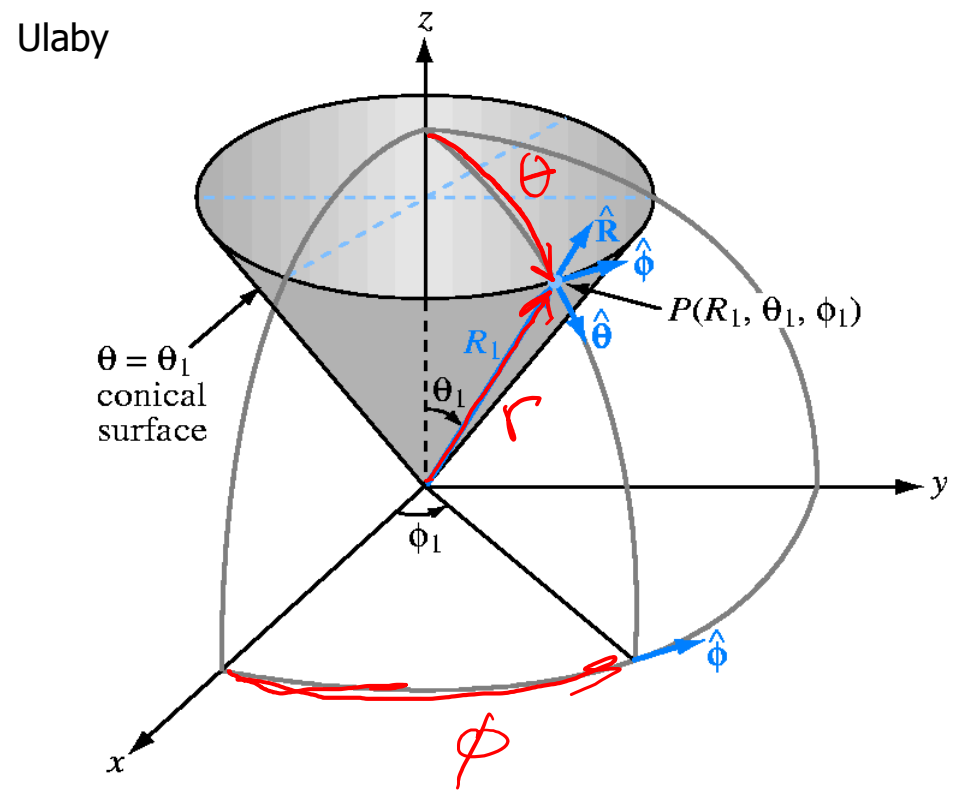


Figure 3-13

Spherical Coordinates

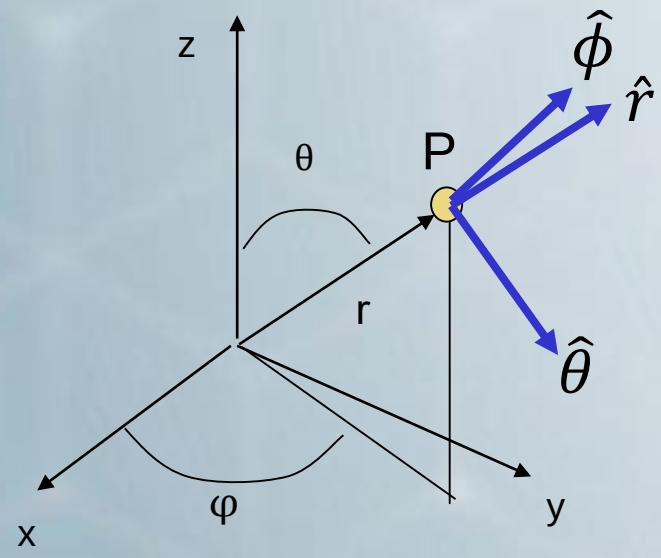
- $\hat{\boldsymbol{r}}$: radial coordinate (length of vector)
- $\hat{\boldsymbol{\theta}}$: angle between vector and +z-axis
- $\hat{\boldsymbol{\phi}}$: angle between vector and x-axis in x-y plane

Position Vector R in Spherical Coordinates:

$$\overrightarrow{OP} = \mathbf{r} = r_1\hat{\mathbf{r}} \text{ for } P(R_1, \theta_1, \phi_1)$$

Field Math

Spherical Coordinates Unit Vector



General Vector A in Spherical Coordinates:

$$\vec{A} = A_r\hat{\boldsymbol{r}} + A_\theta\hat{\boldsymbol{\theta}} + A_\phi\hat{\boldsymbol{\phi}}$$

The Unit Vectors imply :

- $\hat{r}$ ➡ Points in the direction of increasing **r**
- $\hat{\theta}$ ➡ Points in the direction of increasing **θ** (as measured from the +z axis) $0 \leq \theta < \pi$
- $\hat{\phi}$ ➡ Points in the direction of increasing **ϕ** (as measured from the +x axis) $0 \leq \phi < 2\pi$

In spherical coordinates, $\hat{r}$,$\hat{\theta}$,and $\hat{\phi}$ are functions of ϕ and θ

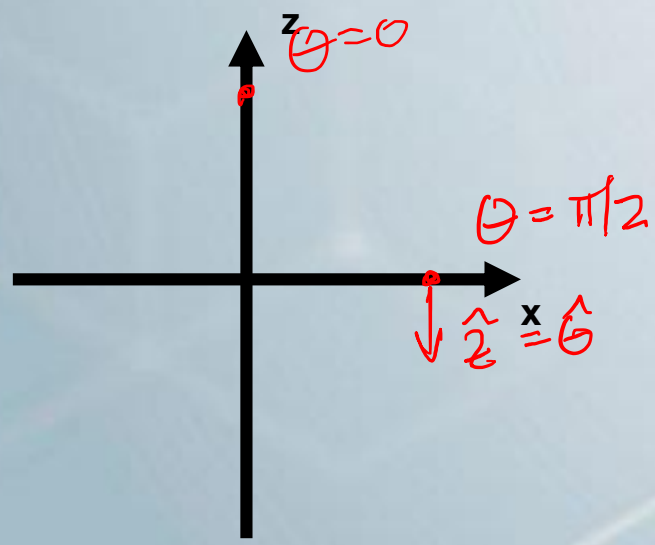
Field Math

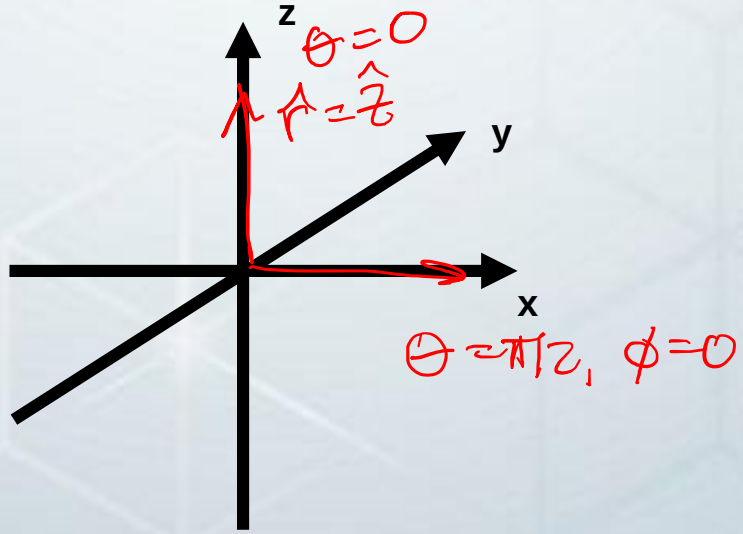
Spherical Coordinates Unit Vector

- What are the unit vectors $\hat{r}$, $\hat{\theta}$, and $\hat{\phi}$ of spherical coordinates in terms of Cartesian Coordinates?
  - What is $\hat{\phi}$?
  - What is $\hat{\theta}$?

$$\hat{r} = (\cos\phi \sin\theta)\hat{x} + (\sin\phi \sin\theta)\hat{y} + \cos\theta\,\hat{z}$$

What is $\hat{r}$?



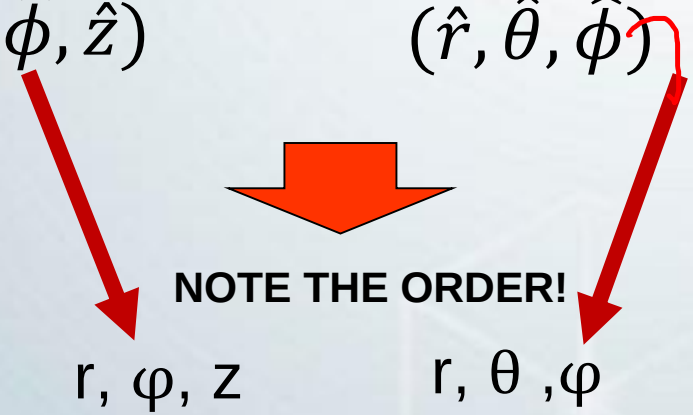


$$\hat{\theta} = (\cos\theta\cos\phi)\hat{x} + (\cos\theta\sin\phi)\hat{y} - \sin\theta\,\hat{z}$$

Field Math

Vector Representation

RECTANGULAR Coordinate System	CYLINDRICAL Coordinate System	SPHERICAL Coordinate System
$(\hat{x}, \hat{y}, \hat{z})$	$(\hat{r}, \hat{\phi}, \hat{z})$	$(\hat{r}, \hat{\theta}, \hat{\phi})$



Note: We do not emphasize transformations between coordinate systems in this course, but it is sometimes necessary for solving problems.

Field Math

Orthogonal Coordinate Systems

$$\vec{A} = A_r\hat{r} + A_\phi\hat{\phi} + A_z\hat{z} = A_x\hat{x} + A_y\hat{y} + A_z\hat{z} = \cdots$$

- Regardless of the orthogonal coordinate system, the following relationships are always true:
  - The dot product of a unit vector with itself is always **1**

$$\hat{r}\cdot\hat{r} = \hat{\phi}\cdot\hat{\phi} = \hat{z}\cdot\hat{z} = \hat{x}\cdot\hat{x} = \hat{y}\cdot\hat{y} = 1$$

  - The dot product of a unit vector with any other unit vector in the coordinate system is always **0** (orthogonality)

$$\hat{r}\cdot\hat{\phi} = \hat{r}\cdot\hat{z} = \hat{\phi}\cdot\hat{z} = \hat{x}\cdot\hat{y} = \hat{y}\cdot\hat{z} = \hat{x}\cdot\hat{z} = 0$$

  - The dot product between two vectors is the sum of the products of the individual vector components in each coordinate

$$\vec{A}\cdot\vec{B} = A_rB_r + A_\theta B_\theta + A_\phi B_\phi = A_xB_x + A_yB_y + A_zB_z = \cdots$$

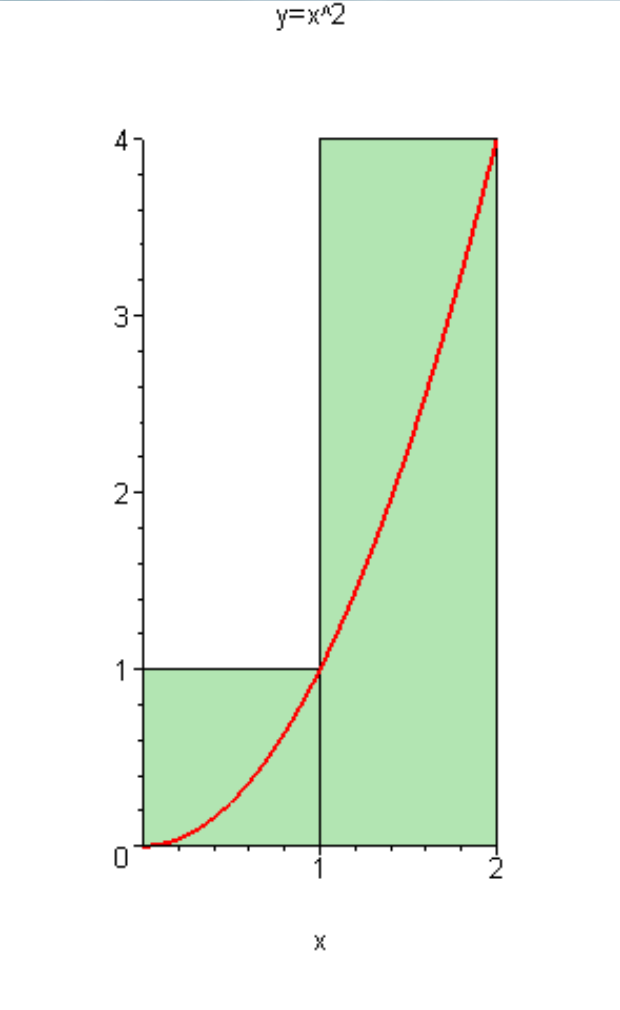
  - The magnitude of a vector is the square root of the dot product of the vector with itself

$$|\vec{A}| = \sqrt{\vec{A}\cdot\vec{A}} = \sqrt{A_r^2 + A_\phi^2 + A_z^2} = \sqrt{A_x^2 + A_y^2 + A_z^2} = \cdots$$

Multidimensional Integrals

Field Math

Differential Calculus



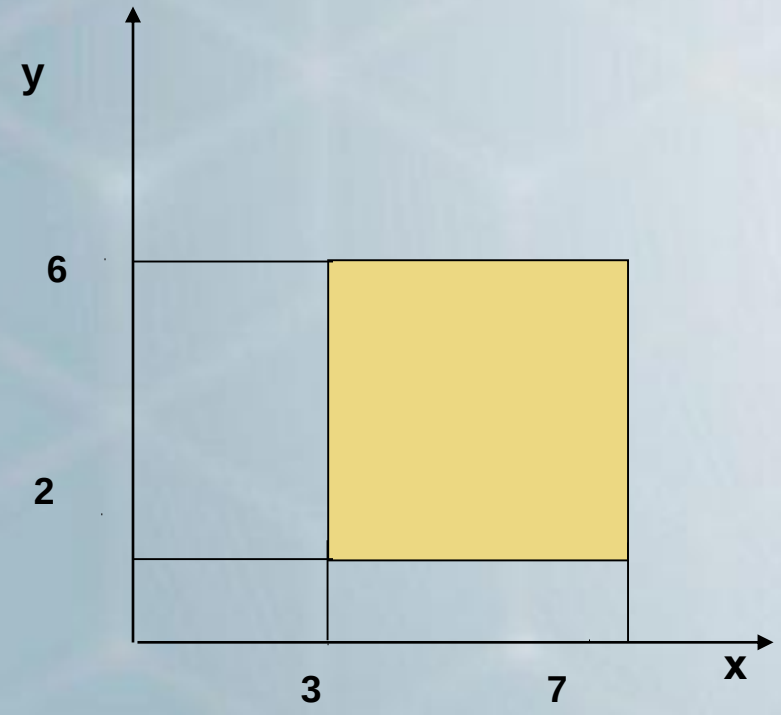
Wikipedia

- Integration = adding up differential elements (infinitesimally small pieces.)
- These differential elements have some specific geometry depending on dimensions/coordinates
- At left, we're integrating infinitesimal slices of y(x)

$$\int f(x)dx$$

Field Math

Differential Calculus



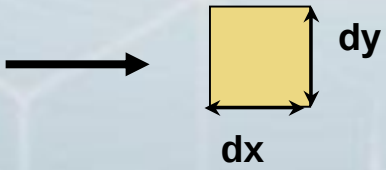
- For 2D Cartesian integrals, we have a square differential element.

$$\text{Area} = \int_2^6 \int_3^7 1 dx dy = 16$$

- Or if we wanted to integrate some function over this area:

$$\text{Area} = \int_2^6 \int_3^7 f(x,y) dx dy = 16$$

Integration over 2 “delta” distances
(2 dimensions – differential area)



Field Math

Differential Calculus

- For integration in three dimensions, we use a triple integral with a differential volume element.

$$\iiint f(\mathbf{r})dV$$

- You can visualize this differential volume element by visualizing the volume that is traced out if you allow a point to "wiggle" a small amount in all three directions specified in the coordinate system.

Field Math
Differential Calculus

- In Cartesian coordinates, the differential element is a cube with size lengths $\mathrm{d}x$, $\mathrm{d}y$ and $\mathrm{d}z$ and volume $\mathrm{d}x\mathrm{d}y\mathrm{d}z$.

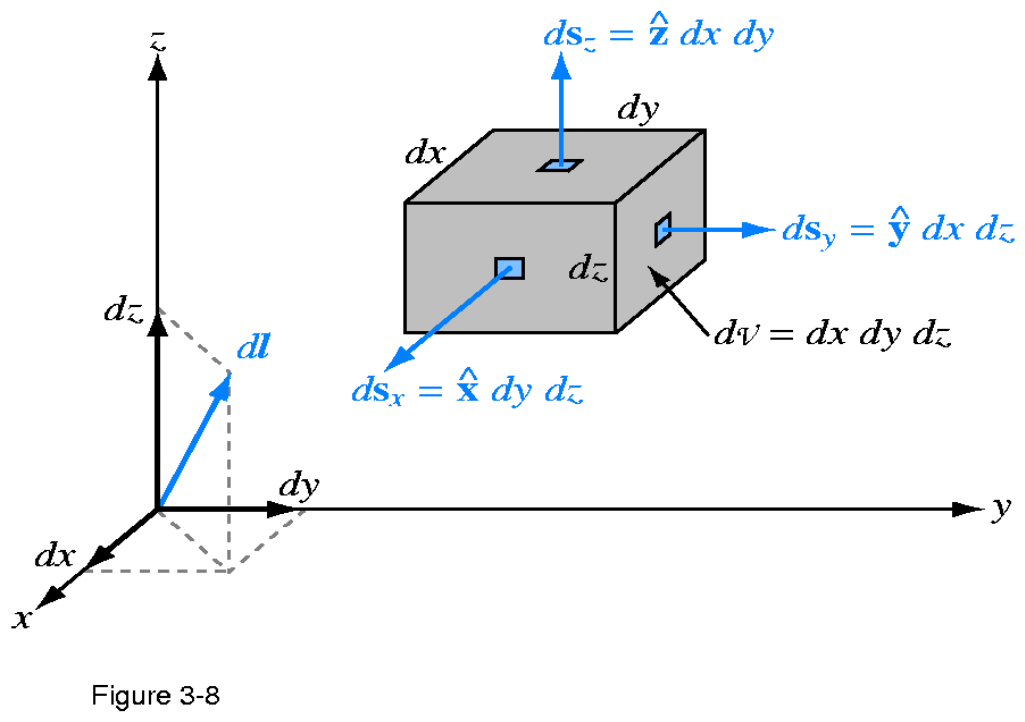


Figure 3-8

$$\iiint f(x,y,z)dxdydz$$

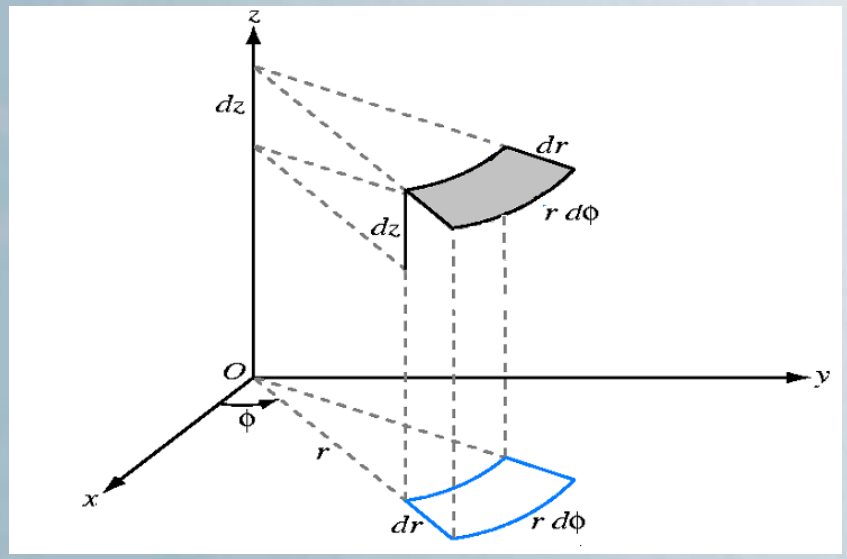
(Recall that we can change the order of integration if we want.)

$$\iiint f(x,y,z)dzdxdy$$

Field Math

Differential Calculus

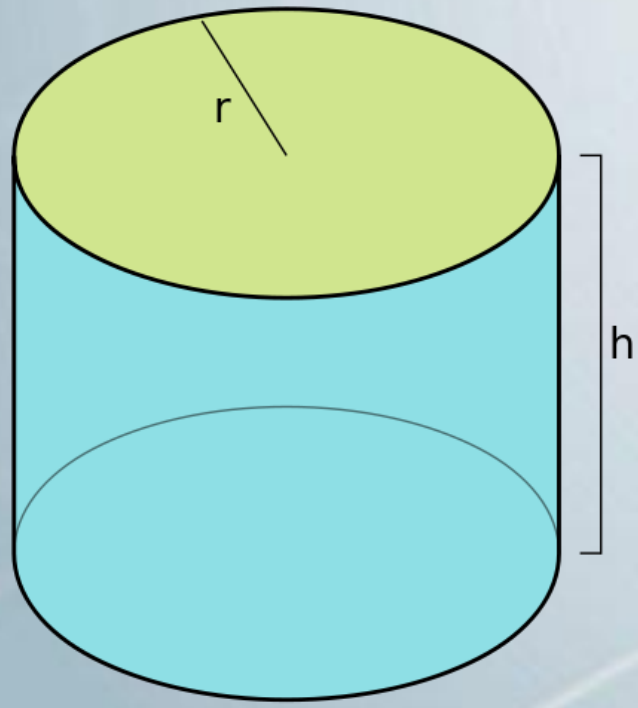
- In cylindrical coordinates, the differential element will be a wedge.



$$\iiint f(r,\phi,z) r dr d\phi dz$$

Field Math
Differential Calculus

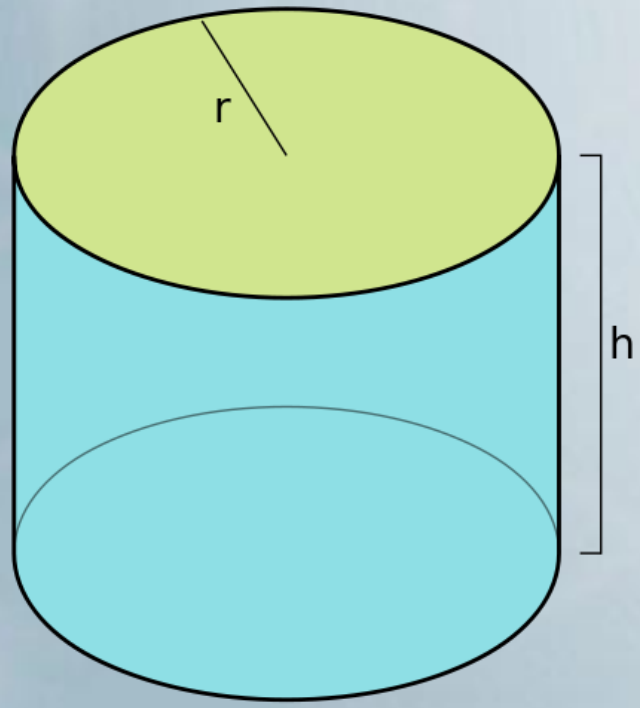
- You can find the volume of this cylinder as follows:



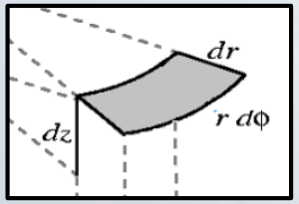
$$\int_0^h \int_0^{2\pi} \int_0^r r dr d\phi dz$$

Field Math
Differential Calculus

- You can find the volume of this cylinder as follows:



$$\int_0^h \int_0^{2\pi} \int_0^r \boxed{r\,dr\,d\phi\,dz}$$



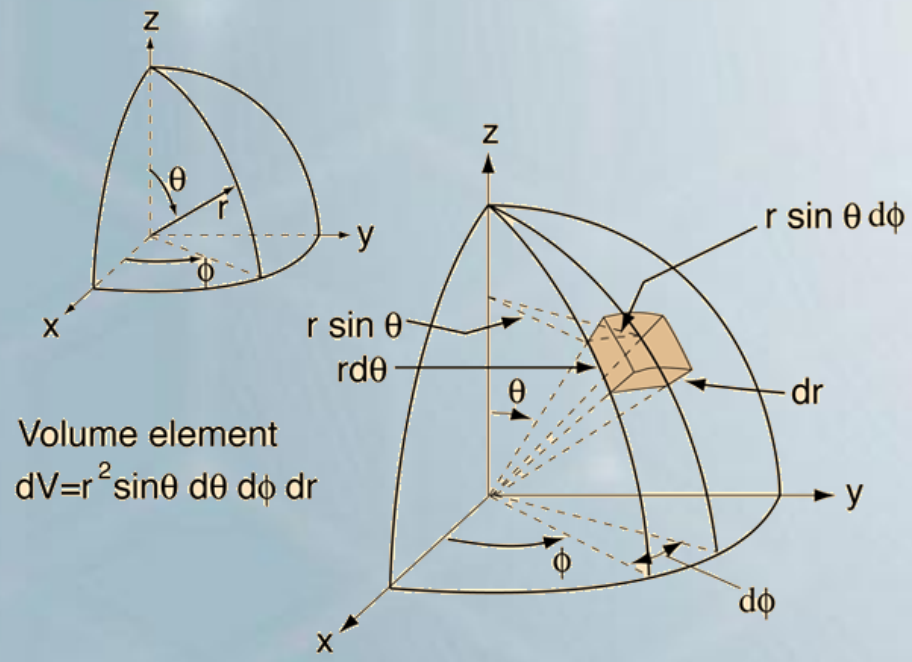
the volume of this

(the volume of the differential wedge looks like that of a rectangular prism. We don't need to worry about curvature due to its infinitesimal size!)

Field Math

Differential Calculus

- In spherical coordinates, two of the wedge dimensions are proportional to r.



$$\iiint f(r,\theta,\phi)r^2 sin\theta dr d\phi d\theta$$

Field Math

Differential Calculus

- Representation of differential lengths **dl** in the three coordinate systems

Rectangular $d\vec{l} = dx \cdot \hat{x} + dy \cdot \hat{y} + dz \cdot \hat{z}$

Cylindrical $d\vec{l} = dr \cdot \hat{r} + rd\phi \cdot \hat{\phi} + dz \cdot \hat{z}$

Spherical $d\vec{l} = dr \cdot \hat{r} + rd\theta \cdot \hat{\theta} + r\sin\theta\, d\phi \cdot \hat{\phi}$

Field Math

Differential Surfaces and Volumes

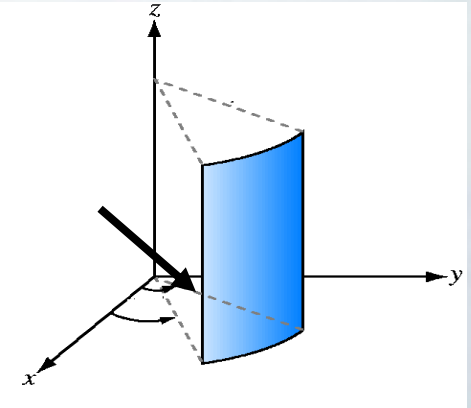
- Example of surface area differentials

$$d\vec{s} = dxdy\hat{z} \quad \text{or} \quad d\vec{s} = rd\phi dz\hat{r}$$

- Representation of differential surface element:

Vector is NORMAL to surface

$$d\vec{s} = rd\phi dz\hat{r}$$



- Differential volume (a scalar)

$$dV = dxdydz$$

Field Math

Differential Surfaces and Volumes

- When doing surface and volume integrals we must consider:
  - What is the right system of coordinates ?
  - What is kept constant?
  - What integral limits do we use?
  - What differential element do we use?

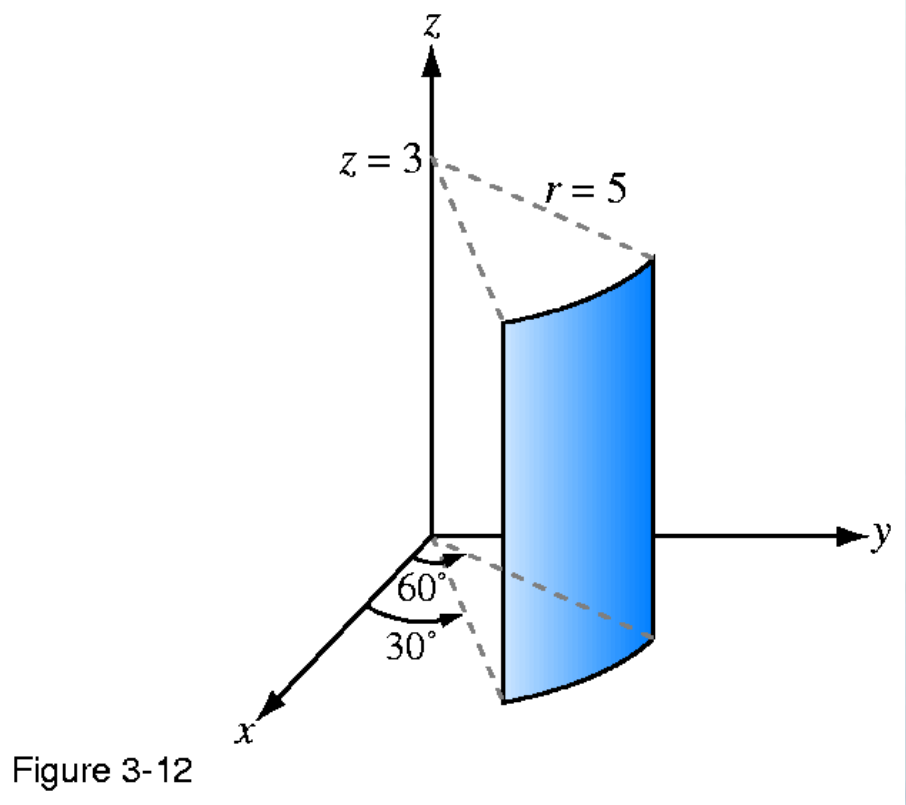


Figure 3-12

Field Math

Differential Surfaces and Volumes

- To visualize a d**S** element:
  - Take the direction that the surface's normal vector points in and hold it constant (i.e. z)
  - Sweep the other two coordinates through all possible values (i.e. x and y)
  - What shape is traced out by this process? This shape is made of all the d**S** elements for the vector you specified.

Field Math

	Cartesian Coordinates	Cylindrical Coordinates	Spherical Coordinates
Coordinate variables	x, y, z	r, ϕ, z	R, θ, ϕ
Vector representation $\mathbf{A} =$	$\hat{\mathbf{x}}A_x + \hat{\mathbf{y}}A_y + \hat{\mathbf{z}}A_z$	$\hat{\mathbf{r}}A_r + \hat{\boldsymbol{\phi}}A_\phi + \hat{\mathbf{z}}A_z$	$\hat{\mathbf{R}}A_R + \hat{\boldsymbol{\theta}}A_\theta + \hat{\boldsymbol{\phi}}A_\phi$
Magnitude of A $\lvert\mathbf{A}\rvert =$	$\sqrt{A_x^2 + A_y^2 + A_z^2}$	$\sqrt{A_r^2 + A_\phi^2 + A_z^2}$	$\sqrt{A_R^2 + A_\theta^2 + A_\phi^2}$
Position vector $\overrightarrow{OP_1} =$	$\hat{\mathbf{x}}x_1 + \hat{\mathbf{y}}y_1 + \hat{\mathbf{z}}z_1$, for $P(x_1, y_1, z_1)$	$\hat{\mathbf{r}}r_1 + \hat{\mathbf{z}}z_1$, for $P(r_1, \phi_1, z_1)$	$\hat{\mathbf{R}}R_1$, for $P(R_1, \theta_1, \phi_1)$
Base vectors properties	$\hat{\mathbf{x}}\cdot\hat{\mathbf{x}} = \hat{\mathbf{y}}\cdot\hat{\mathbf{y}} = \hat{\mathbf{z}}\cdot\hat{\mathbf{z}} = 1$ $\hat{\mathbf{x}}\cdot\hat{\mathbf{y}} = \hat{\mathbf{y}}\cdot\hat{\mathbf{z}} = \hat{\mathbf{z}}\cdot\hat{\mathbf{x}} = 0$ $\hat{\mathbf{x}}\times\hat{\mathbf{y}} = \hat{\mathbf{z}}$ $\hat{\mathbf{y}}\times\hat{\mathbf{z}} = \hat{\mathbf{x}}$ $\hat{\mathbf{z}}\times\hat{\mathbf{x}} = \hat{\mathbf{y}}$	$\hat{\mathbf{r}}\cdot\hat{\mathbf{r}} = \hat{\boldsymbol{\phi}}\cdot\hat{\boldsymbol{\phi}} = \hat{\mathbf{z}}\cdot\hat{\mathbf{z}} = 1$ $\hat{\mathbf{r}}\cdot\hat{\boldsymbol{\phi}} = \hat{\boldsymbol{\phi}}\cdot\hat{\mathbf{z}} = \hat{\mathbf{z}}\cdot\hat{\mathbf{r}} = 0$ $\hat{\mathbf{r}}\times\hat{\boldsymbol{\phi}} = \hat{\mathbf{z}}$ $\hat{\boldsymbol{\phi}}\times\hat{\mathbf{z}} = \hat{\mathbf{r}}$ $\hat{\mathbf{z}}\times\hat{\mathbf{r}} = \hat{\boldsymbol{\phi}}$	$\hat{\mathbf{R}}\cdot\hat{\mathbf{R}} = \hat{\boldsymbol{\theta}}\cdot\hat{\boldsymbol{\theta}} = \hat{\boldsymbol{\phi}}\cdot\hat{\boldsymbol{\phi}} = 1$ $\hat{\mathbf{R}}\cdot\hat{\boldsymbol{\theta}} = \hat{\boldsymbol{\theta}}\cdot\hat{\boldsymbol{\phi}} = \hat{\boldsymbol{\phi}}\cdot\hat{\mathbf{R}} = 0$ $\hat{\mathbf{R}}\times\hat{\boldsymbol{\theta}} = \hat{\boldsymbol{\phi}}$ $\hat{\boldsymbol{\theta}}\times\hat{\boldsymbol{\phi}} = \hat{\mathbf{R}}$ $\hat{\boldsymbol{\phi}}\times\hat{\mathbf{R}} = \hat{\boldsymbol{\theta}}$
Dot product $\mathbf{A}\cdot\mathbf{B} =$	$A_xB_x + A_yB_y + A_zB_z$	$A_rB_r + A_\phi B_\phi + A_zB_z$	$A_RB_R + A_\theta B_\theta + A_\phi B_\phi$
Cross product $\mathbf{A}\times\mathbf{B} =$	$\begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$	$\begin{vmatrix} \hat{\mathbf{r}} & \hat{\boldsymbol{\phi}} & \hat{\mathbf{z}} \\ A_r & A_\phi & A_z \\ B_r & B_\phi & B_z \end{vmatrix}$	$\begin{vmatrix} \hat{\mathbf{R}} & \hat{\boldsymbol{\theta}} & \hat{\boldsymbol{\phi}} \\ A_R & A_\theta & A_\phi \\ B_R & B_\theta & B_\phi \end{vmatrix}$
Differential length $d\mathbf{l} =$	$\hat{\mathbf{x}}\,dx + \hat{\mathbf{y}}\,dy + \hat{\mathbf{z}}\,dz$	$\hat{\mathbf{r}}\,dr + \hat{\boldsymbol{\phi}}r\,d\phi + \hat{\mathbf{z}}\,dz$	$\hat{\mathbf{R}}\,dR + \hat{\boldsymbol{\theta}}R\,d\theta + \hat{\boldsymbol{\phi}}R\sin\theta\,d\phi$
Differential surface areas	$d\mathbf{s}_x = \hat{\mathbf{x}}\,dy\,dz$ $d\mathbf{s}_y = \hat{\mathbf{y}}\,dx\,dz$ $d\mathbf{s}_z = \hat{\mathbf{z}}\,dx\,dy$	$d\mathbf{s}_r = \hat{\mathbf{r}}r\,d\phi\,dz$ $d\mathbf{s}_\phi = \hat{\boldsymbol{\phi}}\,dr\,dz$ $d\mathbf{s}_z = \hat{\mathbf{z}}r\,dr\,d\phi$	$d\mathbf{s}_R = \hat{\mathbf{R}}R^2\sin\theta\,d\theta\,d\phi$ $d\mathbf{s}_\theta = \hat{\boldsymbol{\theta}}R\sin\theta\,dR\,d\phi$ $d\mathbf{s}_\phi = \hat{\boldsymbol{\phi}}R\,dR\,d\theta$
Differential volume $dV =$	$dx\,dy\,dz$	$r\,dr\,d\phi\,dz$	$R^2\sin\theta\,dR\,d\theta\,d\phi$

Field Math

Differential Surfaces and Volumes

Do Lecture 08 Exercise in groups of up to 4.

Wrap-Up

- Exam 1 on Monday
  - Includes material from Lecture 1 – Lecture 7
  - Crib sheet is available on LMS
  - You will be allowed a non-communicating graphic calculator
  - You may bring your own ruler to use with a Smith chart. A Smith chart and a paper ruler will be provided.